



鞍山市千山仪表有限公司

ANSHAN QIANSHAN INSTRUMENT CO., LTD

Expendable Thermocouples · Molten Steel Samplers · Oxygen Probes

ESTABLISHED 1992 · ANSHAN, LIAONING, P.R. CHINA

PRODUCT CATALOGUE



COMPANY PROFILE

A specialist manufacturer of temperature measurement and sampling consumables

Anshan Qianshan Instrument Co., Ltd is a joint-stock enterprise with independent legal-person status, dedicated to the manufacture of expendable thermocouples, molten steel samplers, industrial thermocouples, and paper tubes for thermocouples and samplers in a full range of specifications.

The factory was established in 1992. From an initial registered capital of RMB 500,000 the company has grown to fixed assets of more than RMB 5 million, occupying a site of close to 10,000 m² with a 1,200 m² production hall and a range of modern production and inspection equipment. Output value has exceeded RMB 10 million in each of the last two years, and the company currently serves more than 120 customers.

Qianshan employs 72 staff, of whom more than 30 % are engineering and technical personnel. The company operates under a General Manager responsibility system, with Finance, Quality Control, Sales & Supply and Production departments whose responsibilities are defined in accordance with the company's quality management system.

At a glance

Established	1992
Legal form	Joint-stock enterprise, independent legal person
Site / production hall	Approx. 10,000 m ² / 1,200 m ²
Fixed assets	Over RMB 5 million
Annual output value	Over RMB 10 million (each of the last two years)
Employees	72, of whom over 30 % are engineering and technical staff
Customers served	More than 120
Product range	Expendable thermocouples · molten steel samplers · oxygen probes · thermal analysis cups · compensating cable · instruments · lances, lance heads and cables · paper tubes

Industries served

Basic oxygen and electric arc steelmaking, ladle and secondary refining (LF / RH), continuous casting, foundries (cast iron, cast steel), and non-ferrous melting operations.

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Product range

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01 · EXPENDABLE THERMOCOUPLES

Splash-Proof Expendable Thermocouples

Expendable thermocouples measure the temperature of molten steel and other high-temperature molten metals. Operation is based on the thermoelectric effect: the EMF generated between the hot and cold junctions of the thermocouple pair is converted by the secondary instrument into a temperature reading.

SPLASH-PROOF DESIGN

The splash-proof type is intended for cast steel, cast copper, cast aluminium and cast iron. The protective head shields the operator from molten metal spatter during immersion. Each probe can be used repeatedly for 8–10 measurements, which substantially reduces the cost per measurement compared with single-use probes.



Splash-proof probes with steel protective cap



Splash-proof probes with aluminium protective cap



Splash-proof expendable thermocouples with retaining eyelet, suitable for repeated immersion

Key features

- Reusable for 8–10 measurements per probe
- Splash protection during immersion improves operator safety
- Suitable for cast steel, cast copper, cast aluminium and cast iron
- Available in graduation types S, R, B and W (Type C)
- Lower cost per measurement than single-use probes

02 · EXPENDABLE THERMOCOUPLES

Series 602 and Series 604

The 602 and 604 series are single-use immersion probes for molten steel and molten iron. Both series are supplied in all four graduation types, with the contact block dimension and head form matched to the customer's measuring lance.

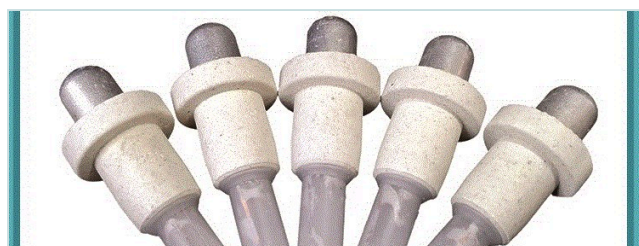
SERIES 602 — KS-602 · KR-602 · KB-602 · KW-602



KS-602 probe assemblies on paper tube



602 measuring heads, steel cap



602 measuring heads with quartz tube



KW-602 probe assemblies

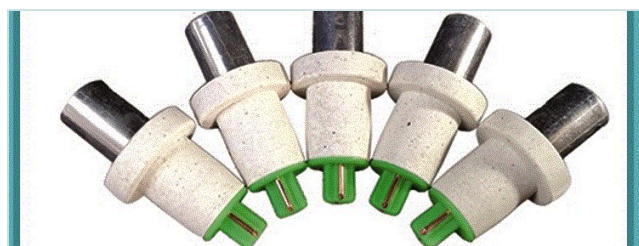
SERIES 604 — KS-604 · KR-604 · KB-604 · KW-604



KS-604 probe assemblies on paper tube



KB-604 probe assemblies



604 measuring heads, colour-coded contact



604 measuring heads, aluminium cap

03 · EXPENDABLE THERMOCOUPLES

Graduation Type R — Models HR and OR

Graduation type R (Pt-13 % Rh / Pt) is supplied in four models — KR-602, KR-604, HR and OR. HR and OR are distinct contact block designs for customers whose measuring lances require them; they are identical in thermoelectric performance to the KR-602 and KR-604 probes.

Model	Graduation	Element combination	Head form
KR-602	R	Pt-13 % Rh / Pt	602 contact block
KR-604	R	Pt-13 % Rh / Pt	604 contact block
HR	R	Pt-13 % Rh / Pt	HR contact block
OR	R	Pt-13 % Rh / Pt	OR contact block



Model HR measuring heads



Model OR measuring heads

Protective caps

All expendable thermocouple series can be supplied with copper, aluminium or steel protective caps. The cap material is selected according to the melt and the immersion conditions.



Probes with copper protective cap



Probes with aluminium protective cap

04 · EXPENDABLE THERMOCOUPLES

Operating instructions and technical data

Operating instructions

1. Select the paper tube length and the measuring lance according to the vessel and the measuring range.
2. Fit the expendable thermocouple onto the measuring lance. The secondary instrument should return to zero, confirming good electrical contact before measurement.
3. Immerse the probe 300–400 mm below the surface of the melt. Do not touch the vessel wall or the slag layer. Withdraw the lance as soon as the instrument has taken the reading; total immersion time must not exceed 5 seconds, otherwise the measuring lance may be damaged.
4. After the lance is withdrawn from the furnace, remove the used probe and fit a new one. Allow a few minutes before the next measurement. Continuous measuring and changing without a pause will cause temperature scatter.

Specifications and performance

Element combination	Model	Type	Tolerance	Service temperature		Measuring time
				Optimum range	Upper limit	
Pt-30 % Rh / Pt-6 % Rh	KB-602P	B	±5 °C	1500–1700 °C	1750 °C	4–6 s
Pt-10 % Rh / Pt	KS-602P	S	±5 °C	1400–1600 °C	1650 °C	4–6 s
Pt-13 % Rh / Pt	KR-602P	R	±5 °C	1400–1600 °C	1650 °C	4–6 s
W-Re 3 / W-Re 25	KW-602P	W (Type C)	±7 °C	1500–1700 °C	1800 °C	4–6 s

Models shown are the 602 series. The same element combinations are available in the 604, HR and OR models. Graduation W (W-Re 3 / W-Re 25) corresponds to the international designation Type C.

Technical data

Maximum wire temperature — type W / Type C (W-Re 3/25)	1820 °C
Maximum wire temperature — type B (Pt-30 % Rh / Pt-6 % Rh)	1800 °C
Maximum wire temperature — types S and R	1760 °C
Tolerance — standard class (P) / premium class (J)	±5 °C / ±2 °C
Wire accuracy at the palladium point (1554 °C), class P / class J	±3 °C / ±1.5 °C
Paper tube length	300–1800 mm, made to order
Protective cap material	Steel or aluminium
Measurement success rate	≥ 98 %
Response time	≤ 5 s

Transport and storage

Handle with care during loading and unloading. Protect from moisture in transit. Store in a dry warehouse, in the original cartons, at a relative humidity not exceeding 80 %, with good air circulation and free from corrosive gases.

Molten Steel Samplers

Molten steel samplers take a fully formed steel sample for spectrometric analysis. They are simple in construction, convenient in use and save considerable time compared with ladle sampling, with a sampling success rate above 98 % and dense, sound samples free from porosity.

Samplers are produced in single-chamber and dual-chamber designs. Types are available both for refining, ladle and tundish applications where no deoxidation is required, and for BOF, EAF and open-hearth applications where the melt must be deoxidised. Sampler styles can be manufactured to customer drawing.

DEOXIDISER VARIANTS

- KQY** — with deoxidiser, for sampling steel of high oxygen content.
- KQN** — without deoxidiser, for sampling steel of low oxygen content.

Sample dimensions

Sample	Diameter	Thickness / length	Mass
Disc sample (cup mould)	Ø 38 mm	17 mm	150 g
Pin sample (quartz tube)	Ø 6 mm	14-35 mm	31-77 g

Sampling lances

KQ-2M (2 m) · KQ-2.5M (2.5 m) · KQ-3M (3 m) · KQ-3.5M (3.5 m). Please state the required lance length when ordering. For nitrogen, hydrogen and oxygen analysis, a vacuum glass sampling tube of Ø 3 mm × 200 mm is used.

Operation

1. Fit the sampler onto the dedicated sampling lance.
2. Immerse the sampler into the melt at an angle of 60–65°, to a depth of at least 350 mm, for 4–6 seconds.
3. After withdrawal from the furnace or ladle, tap the sampler lightly on the floor to release the sample.

Precautions

1. Samplers must be protected from moisture during storage.
2. Sample only when the steel temperature is above 1500 °C.
3. Avoid heavy shock, which may fracture the quartz tube.
4. Bake samplers at 60 °C for 10 hours before use.
5. Observe the specified immersion depth and dwell time.

Advantages

1. Simple operation, easily mastered by the operator.
2. Furnace-side samplers require immersion to only 300 mm below the surface for 5–8 seconds. The sample is representative and is not affected by operator technique as ladle-spoon sampling is.
3. Casting-platform samplers need only 2–3 seconds of contact between the quartz tube tip and the pouring stream. The stream does not have to be interrupted, which is safe and reliable.
4. Process samplers and finished-product rimming-steel samplers contain a measured 0.5 g aluminium deoxidiser, so results are not affected by variations in the aluminium addition.
5. Once cooled, the sample is labelled and sent to the laboratory for grinding and analysis — about two minutes faster than a conventional mould sample.
6. Because samples are labelled directly, the steel tag formerly used for identification is no longer required.

Thermal Analysis Cups and Sampling Spoons

Thermal analysis cups consist of a carefully engineered cup body and a high-precision quartz-tube thermocouple.



Sampler assemblies on paper tube



Sampling spoon

Principle and benefits

Thermal analysis is widely used for furnace-side control of hot metal composition because it is fast, simple, reliable and inexpensive. When hot metal is poured into the cup it solidifies progressively; the temperature arrests on the cooling curve are used to determine and calculate CEL, C and Si.

Application

Thermal analysis of white iron, grey iron and ductile iron. Used together with a furnace-side hot metal analyser or carbon analyser, the carbon cup determines carbon content C %, silicon content Si % and carbon equivalent CEL within three minutes. This allows the mechanical and physical properties of the hot metal to be estimated and the composition to be adjusted before pouring.



Thermal analysis cup and mould body



Sampling spoon with refractory lining

Measuring range and accuracy

Thermocouple	Type K (NiCr-NiSi) to IEC 584, up to 1370 °C
Carbon, C	2.1-4.2 %
Silicon, Si	0-20 %
Carbon equivalent, CEL	2.5-5.0 %
Temperature accuracy	≤ ±1 °C
Accuracy — CEL / C / Si	±0.05 % / ±0.05 % / ±0.1 %

07 · OXYGEN PROBES

Oxygen Probes

The oxygen probe measures the temperature and the oxygen activity of molten steel in a single immersion. It is a single-use consumable and an essential process-control tool in modern steelmaking.

Description

The probe is fitted to a 1.2 m refractory paper tube and connected through the measuring lance to a dedicated industrial oxygen determination instrument. The complete measurement is obtained within 8 seconds and is applicable to steel and iron melts across the range of steelmaking operations.

As the production of rimming steel, semi-killed steel and aluminium-killed continuous-cast steel has expanded, and as secondary refining has become widespread, fast and direct knowledge of the oxygen content of the melt has become essential — to control the steelmaking operation, raise steel quality, increase yield, shorten heating and laboratory analysis times and reduce consumption and cost. Solid-electrolyte concentration-cell oxygen measurement was developed to meet these requirements.

The probe offers stable and repeatable performance with good thermal shock resistance. It is used at LF and RH refining stations to measure the oxygen activity of the arriving heat and during treatment, so that the deoxidiser addition can be determined accurately, treatment time shortened and steel quality improved.

Principle

The probe uses a zirconia solid electrolyte to measure the oxygen content of the melt. On immersion, electrode reactions establish different equilibrium electrode potentials at the electrolyte interfaces. The probe consists of a half-cell and a thermocouple, so that steel temperature and oxygen content are measured simultaneously. A reference electrode of known oxygen partial pressure and the melt under test are connected by the oxygen-ion-conducting solid electrolyte, forming an oxygen concentration cell. From the measured temperature and oxygen EMF, the oxygen content of the steel is calculated.

CELL CHAIN

electrode / $p_{O_2}(\text{ref.})$ // electrolyte // $p_{O_2}(\text{steel})$ / electrode
 $\text{Mo} / \text{Cr} + \text{Cr}_2\text{O}_3$ // $\text{ZrO}_2(\text{MgO})$ // $[\text{O}]\text{Fe} / \text{Fe}$

The potential difference follows the Nernst equation. Since the oxygen partial pressure of the solid electrolyte reference is known and R and F are constants, measuring the steel temperature T and the oxygen EMF gives the oxygen activity of the melt directly.

In the production of low-carbon aluminium-killed steel the deoxidation reaction is $2\text{Al} + 3[\text{O}] \rightarrow \text{Al}_2\text{O}_3$. After deoxidation the aluminium and oxygen in the melt reach equilibrium, so that the acid-soluble aluminium content can be derived from the measured free oxygen.



Oxygen probes for in-furnace and in-ladle measurement

08 · OXYGEN PROBES

Specifications and operation

Principal technical data by measuring location

Location	Reference material	Range	Accuracy	Success rate	Response
In furnace	Mo / MoO ₂ , Cr / Cr ₂ O ₃	50–1200 ppm	10 %	Oxygen ≥ 90 % Temperature ≥ 98 %	≤ 8 s
In ladle	Cr / Cr ₂ O ₃	1–100 ppm	≥ 10 ppm: error 10 % ≤ 10 ppm: error ±1 ppm	Temperature ≥ 98 %	≤ 8 s

Model and specification

Model	ZCDY-(S / B / R)
Dimensions	1 × 38 × 1200 mm
Temperature range, type S (Pt-10 % Rh / Pt)	400–1760 °C
Temperature range, type R (Pt-13 % Rh / Pt)	400–1760 °C
Temperature range, type B (Pt-30 % Rh / Pt-6 % Rh)	600–1800 °C
Oxygen EMF	–300 to +300 mV
Oxygen activity range	1–1500 ppm
Oxygen tolerance, ≤ 10 ppm	±1 ppm
Oxygen tolerance, > 10 ppm	±5 % of reading
Output EMF consistency	≤ 2 mV
Electrical requirement	Continuity, correct polarity
Aluminium content measuring range	0.01–0.10 %
Response time — oxygen cell	3–6 s
Response time — thermocouple	2–5 s
Response time — complete measurement	3–8 s
Thermocouple accuracy, 1200–1760 °C	±3 °C
Paper tube length	1.2 m, customisable

Linear measuring ranges are stated to IEC 584 / IPTS-68.

Operation

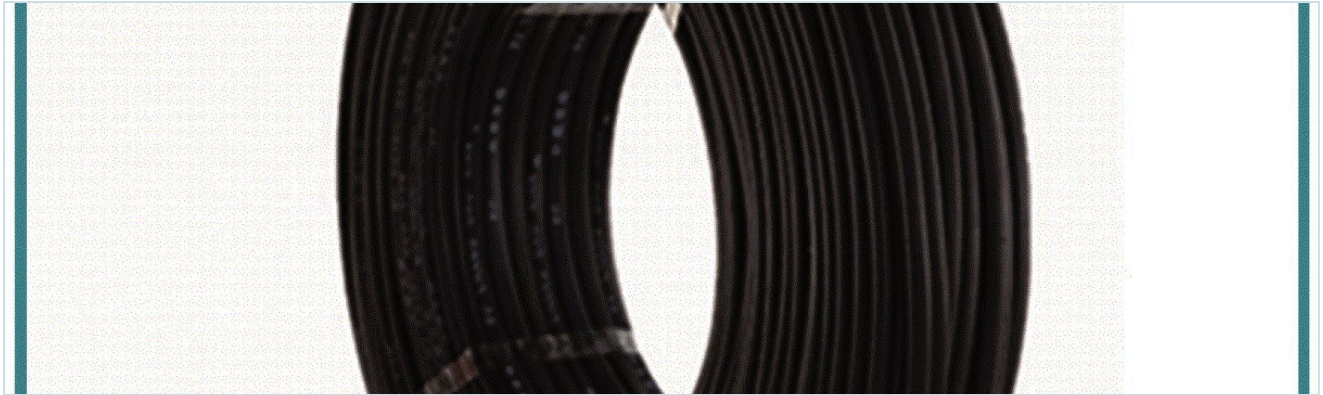
1. Inspect the probe, the measuring lance and the instrument system carefully and confirm that all are in good condition.
2. Fit the oxygen probe onto the measuring lance. The instrument pointer or digital display returning to zero indicates good continuity; measurement may then proceed.
3. Immersion time in the melt must not exceed 10 seconds, otherwise the measuring lance may be damaged.

09 · ACCESSORIES

Compensating Cable

Compensating cable is a pair of insulated conductors which, over a defined temperature range including ambient, produce the same nominal thermal EMF as the thermocouple with which they are matched.

They connect the thermocouple to the measuring device and compensate the error that would otherwise arise from temperature variation at the junction.



Compensating cable

Function

The cable extends the thermoelectrodes, effectively relocating the cold junction of the thermocouple, and together with the indicating instrument forms the complete temperature measuring loop.

Standard

Manufactured to the national standard equivalent to IEC 584-3, *Thermocouples — Part 3: Extension and compensating cables*.

Applications

Used in temperature measuring installations of all kinds, and applied widely in the nuclear power, petroleum, chemical, metallurgical and electric power industries.

ORDERING INFORMATION

Please state the matched thermocouple graduation type, the insulation and sheath material, the number of pairs, whether screening is required, and the length. Cable is supplied in coils to order.

10 · INSTRUMENTS

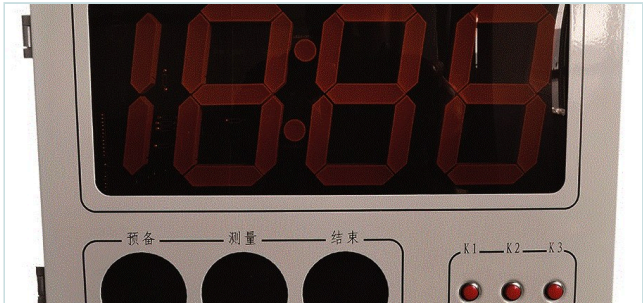
Instruments and Measuring Lances

Temperature indicator — basic performance

Temperature range	300–1400 °C · 600–2000 °C
Accuracy	±0.1 % or ±1 °C, whichever is greater (at 23 ± 5 °C)
Repeatability	±0.1 % or ±0.5 °C, whichever is greater (at 23 ± 5 °C)
Emissivity	0.10–1.00 (factory set 0.95)
Ambient temperature	–10 to 60 °C, without cooling
Analogue output	4–20 mA, two-wire, max. load 750 Ω
Sighting	Laser (Class 2, max. output 1 mW, 650 nm)
Dimensions	Ø 45 mm × 120 mm



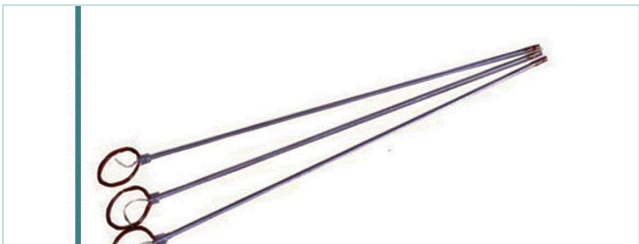
Furnace-side hot metal analyser with printer



Digital molten steel temperature indicator, large display

Measuring lances

Measuring lances are supplied in fixed and portable forms and in a range of lengths to suit the vessel and the working position. The portable lance integrates the indicator into the handle for furnace-side use.



Measuring lances

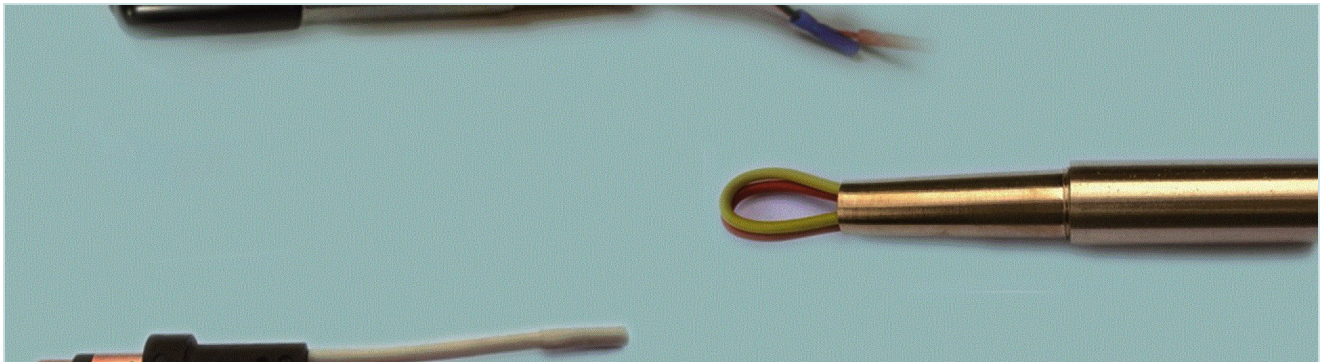


Portable measuring lance with integrated indicator

11 · ACCESSORIES

Lance Heads (Contact Blocks)

The lance head is the connector between the expendable thermocouple and the measuring instrument. It suits all series of Qianshan expendable thermocouples and provides a secure, low-resistance contact under furnace-side conditions.



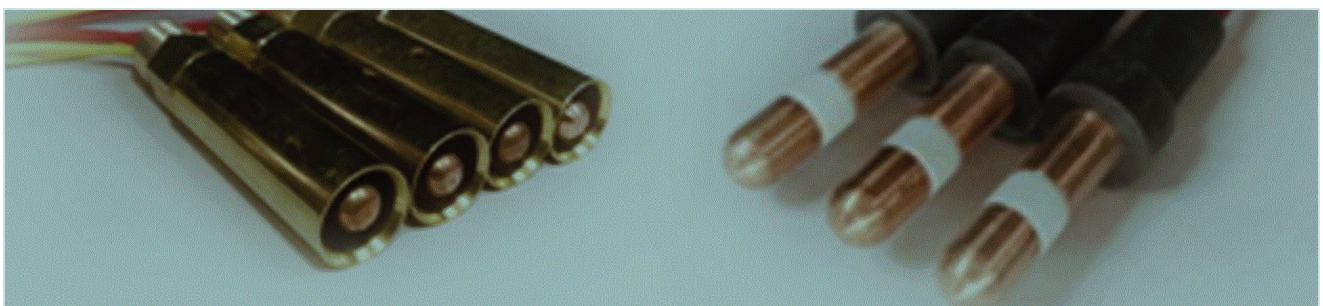
Lance heads and contact assemblies

Construction

- Brass housing; contact pin available in brass or copper.
- High-grade insulating material that does not deform at elevated temperature.
- Polymer insulation rated for continuous service at 200 °C, integrally moulded for high positional accuracy.
- Lead-out wires attached by welding rather than soft soldering, eliminating a third thermoelectric element in the circuit and improving measuring accuracy.
- Compensating cable of the corresponding graduation type, correctly polarised.

Application

Used mainly for fast temperature measurement in the metallurgical and foundry industries. The lance head connects the miniature thermocouple, the measuring lance, the compensating cable and the secondary instrument into a complete measuring loop and transmits the temperature signal.

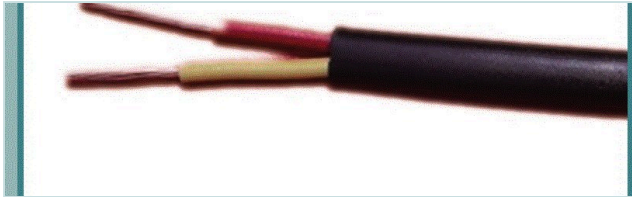


Contact blocks with welded lead-out wires

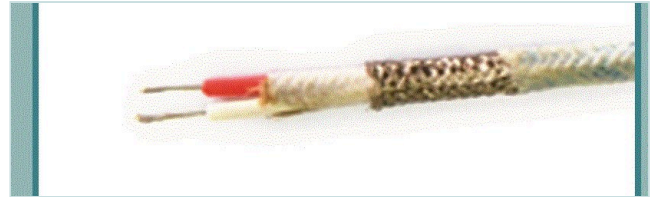
12 · ACCESSORIES

Lance Cables

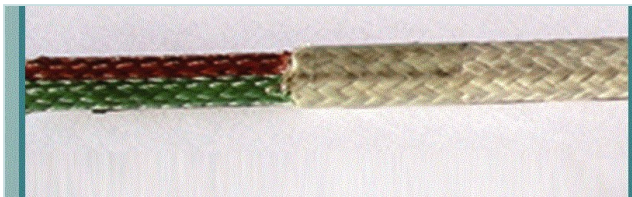
Lance cables connect the lance head to the secondary instrument. They are supplied in a range of insulation and sheath materials — PVC, fibreglass braid, silicone and screened constructions — selected according to the ambient temperature, mechanical duty and electrical interference at the measuring position.



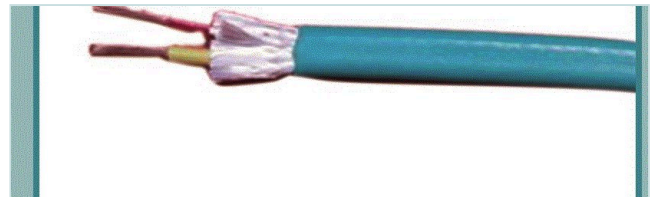
PVC insulated and sheathed



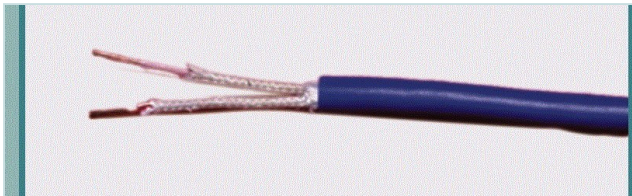
Screened, fibreglass braided



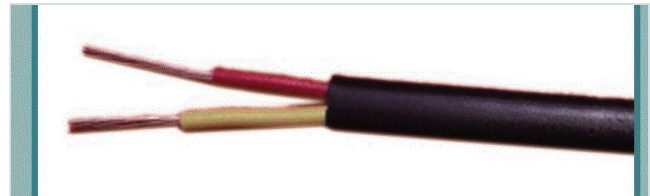
Fibreglass braided, high temperature



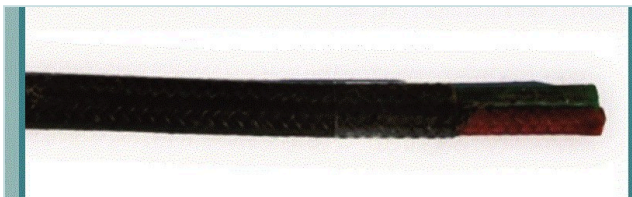
Screened, PVC sheathed



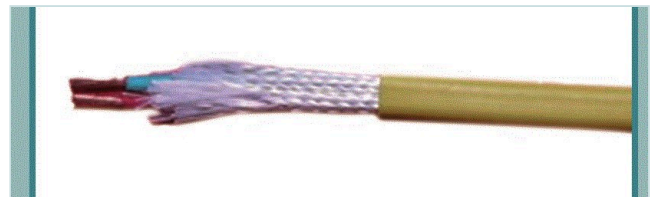
Screened, PVC sheathed



PVC insulated and sheathed



Fibreglass braided



Screened, high temperature

ORDERING INFORMATION

Please state the thermocouple graduation type, conductor cross-section, insulation and sheath material, screening requirement and length.



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